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# Unit 4

# How to write DISCUSSION

Teaching Team of Ethics and Scientific Writing Technique Course

Faculty of Engineering
Universitas Gadjah Mada, 2025




Source: Facebook of High impact PhD memes

# The Shape of a Research Article

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Prepare Figures/Graphs ✓

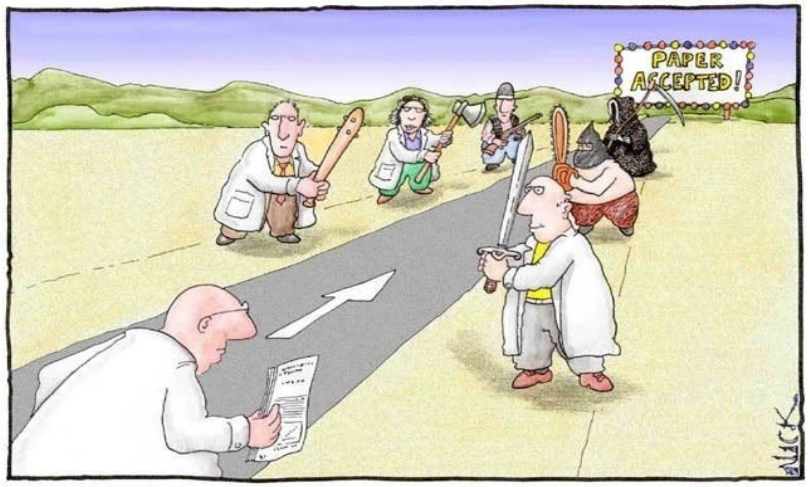
Prepare Results/Discussion

Prepare Methodology ✓

Prepare Introduction ✓

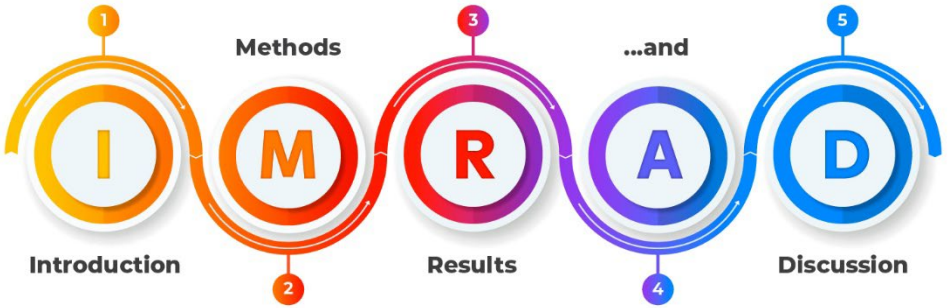
Prepare Abstract & Title

Order of our work when we prepare our manuscript



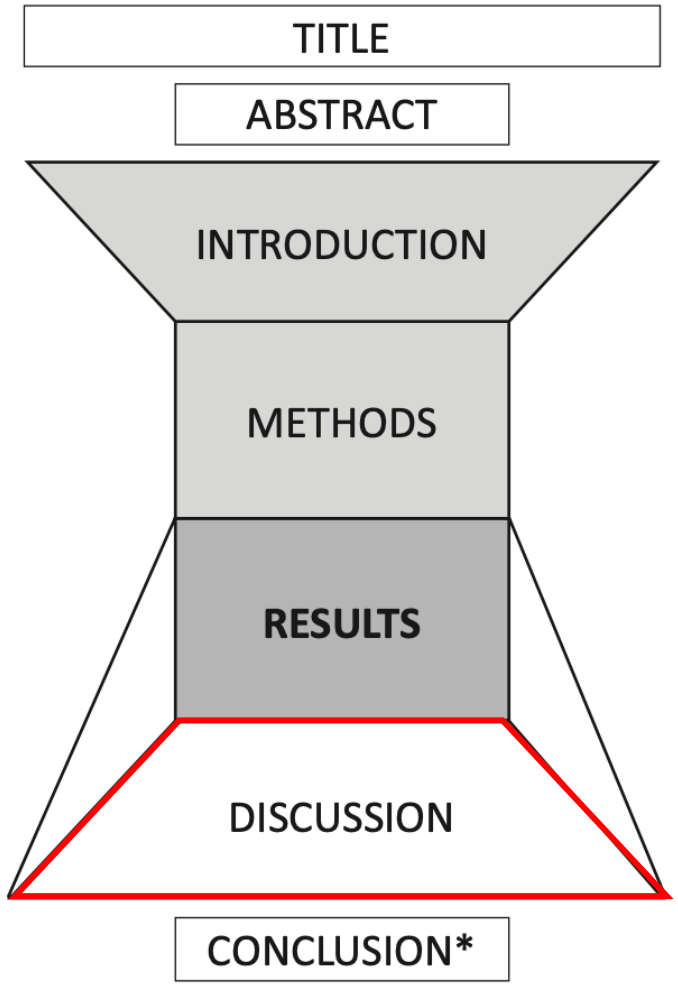

Most scientists regarded the new streamlined peer-review process as 'quite an improvement.'

www.weirdscience.ca



# The Shape of a Research Article: Discussion

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# Contents

1. How to write the Discussion
2. Building your own model
3. Testing and Adjusting the Basic Generic Model
4. Useful Words and Phrases, Modal Verbs

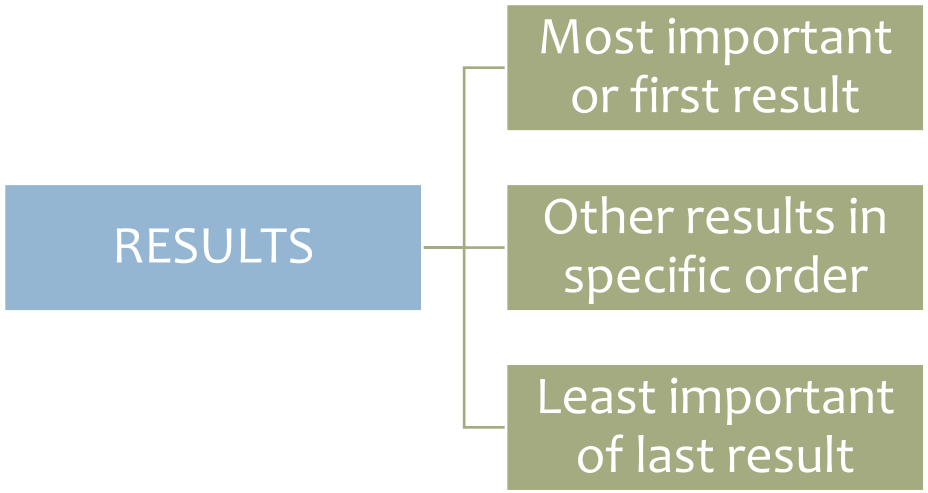
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# Results vs Discussion

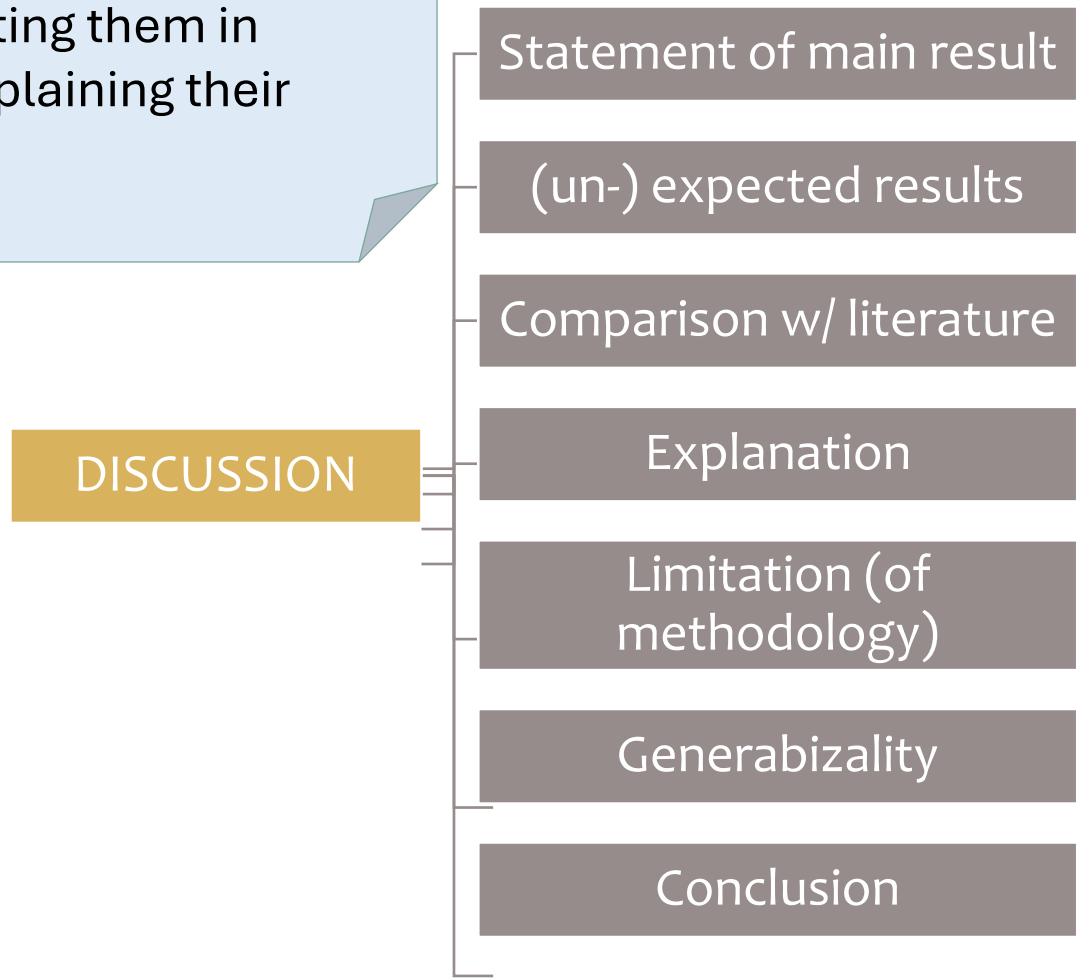
# Review: The difference between Results and Discussion

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The results section objectively reports your findings as they are; no speculations on why you found the results.




The discussion section interprets the results, putting them in context, and explaining their importance.



# Review: The difference between Results and Discussion

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| Results section | Discussion section |
| --- | --- |
| This section answers the question “What?” about your research | This section answers the question “So what?” about your research |
| Describes the experiments conducted before writing the paper | Summarizes and interprets the significance of the key findings |
| States the results but does not interpret them | Interprets the results but does not re-state them |
| Includes only data that will be important to the Discussion section | Does not present new results, so do not make statements that your results cannot support |
| Uses the past simple | Uses both past and present tense as needed |
| It may include non-textual elements such as tables, pictures, and photographs | Use only text, although you may also refer to non-text elements |

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

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# Generic Discussion Model

# Types of scientific contributions (before UTS)

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- Working on **old problem** with **new method** **(*improvement-type* research, contribution in method)**

- Implementing **old method** to **new problem** **(contribution in problem)**

- Implementing **new method** to **new problem**

- Proofing a theoretical model (computational) with real implementation in hardware.

- Development of **low-cost** method / technique / hardware / system with **equal performance** of existing method / technique / hardware / system



# Reviewing the Generic Introduction Model (before UTS)

| GENERIC INTRODUCTION MODEL (FROM Unit 1) | |
|---|---|
| Block 1 | ESTABLISH THE IMPORTANCE OF THE TOPIC/FIELD<br>PROVIDE BACKGROUND FACTUAL INFORMATION<br>PRESENT THE GENERAL PROBLEM AREA/CURRENT RESEARCH FOCUS |
| Block 2 | PRESENT PREVIOUS AND/OR CURRENT RESEARCH AND CONTRIBUTIONS: the research 'map' |
| Block 3 | LOCATE A GAP IN THE RESEARCH<br>DESCRIBE THE PROBLEM YOU WILL ADDRESS<br>PRESENT YOUR MOTIVATION AND/OR HYPOTHESIS<br>IDENTIFY RESEARCH OPPORTUNITY |
| Block 4 | DESCRIBE THE PRESENT PAPER, sometimes mentioning aims/results/methods/conclusions, and often including 'happy' ☺ words |




The Introduction and the Discussion mirror each other.

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

# Before Planning the Discussion

Let's step back from the study and considering its main value or contribution.
For example:

- ❑ Has the study obtained identical or similar results to other studies but uses a modified or new method that is better than existing methods? In that case, the main contribution may be the method.
- ❑ Has the study obtained better (e.g. more accurate) results than other studies? In that case, the main contribution may be the results themselves.
- ❑ Is the study a game-changer, i.e. is it setting a new direction for research/invalidating previous work? In that case, the main contribution of the study may be its impact on the literature/research world.
- ❑ Has the study identified or created new or extended applications? In that case, the main contribution may be its impact on industry/the real world.

https://science-publisher.org/how-to-write-the-results-and-discussion-sections-in-a-research-paper/

# Present Your Contribution in the Discussion Part



Your hypotheses/proposals

Your results

Results from literaturs

Results by other scientists

Prior beliefs/knowledge in your field

General consensus in your field

Don’t be shy as a new researcher to present your own contribution. Yet, please be humble and objective too

Picture from: https://tobeme.com.au/

- ❑ Discussion part is the place where you present your own idea / hypotheses that you want to deliver to the public (based on your results) → many students (first timers) treat discussion part just by playing safe (“normative conformity”)
- ❑ Control your level of claim: Don’t be over confident but don’t underplay your results.
- ❑ We use simple present tense to show **high level of confidence**, and we use simple past tense to **lower our level of confidence**.

# Build Discussion Model

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**An assessment of post V2D sequelae in disease survivors**

***Discussion***

***1*** *To our knowledge, the study reported here is the largest study of patients who have survived the viral disease V2D, and provides a key contribution to the understanding of post-V2D health issues following infection.* ***2*** *Our study is the first to record a wide range of post-V2D complications, and provides clear evidence that survivors who experience a given symptom during V2D infection often present with a closely-related symptom within 300 days following infection.* ***3*** *This was particularly true in relation to some specific complications, such as hearing loss (Fig. 1).* ***4*** *A key finding was that some post-V2D health issues appear to be age-related: infants below the age of 2 experienced post-V2D cardiac complications far less frequently than adults irrespective of the severity of such complications during the acute stage (Figs. 2a and 2b).* ***5*** *This has clear implications for post-V2D-related public health management, and suggests that when resources are limited, cardiac screening priority could be given to adults.*

***6*** *Some questions remain; for example, anecdotal evidence collected from patients during the study period suggested that there was a positive correlation between viral load at the acute stage and the severity of post-V2D health issues.* ***7*** *However, this was not addressed formally during the study.* ***8*** *Although many studies have quantified viral load during acute V2D[17,22,27,33], a direct comparison with the post-viral period would provide valuable data to support such a correlation.*

***9*** *V2D RNA is known to persist in the bodies of survivors long after disease onset[30,42], and the results of our study provide compelling evidence that follow-up of all convalescent patients is warranted for at least 10 months after discharge from hospital.* ***10*** *In addition, the study provides evidence-based guidance for pragmatic public health recommendations that include age-related follow-up and treatment of post-V2D complications.*

# Key of Discussion Model

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**In Sentence 1** *To our knowledge, the study reported here is the largest study of patients who have survived the viral disease V2D, and provides a key contribution to the understanding of post-V2D health issues following infection.* **the writer explicitly identifies the contribution of the study in relation to the existing literature and knowledge.**

**In Sentence 2** *Our study is the first to record a wide range of post-V2D complications, and provides clear evidence that survivors who experience a given symptom during V2D infection often present with a closely-related symptom within 300 days following infection.* **the writer identifies the specific novel feature of the study in relation to the existing literature, and summarises the results.**

**In Sentences 3 and 4** *This was particularly true in relation to some specific complications, such as hearing loss (Fig. 1). A key finding was that some post-V2D health issues appear to be age-related: infants below the age of 2 experienced post-V2D cardiac complications far less frequently than adults irrespective of the severity of such complications during the acute stage (Figs. 2a and 2b).* **the writer revisits the results and comments on them** *(particularly true…a key finding was…appear to be…far less)*.

# Key of Discussion Model

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**In Sentence 5** *This has clear implications for post-V2D-related public health management, and suggests that when resources are limited, cardiac screening priority could be given to adults.* **the writer mentions a potential application arising from the results.**

**In Sentences 6, 7 and 8** *Some questions remain; for example, anecdotal evidence collected from patients during the study period suggested that there was a positive correlation between viral load at the acute stage and the severity of post-V2D health issues. However, this was not addressed formally during the study. Although many studies have quantified viral load during acute V2D*[17,22,27,33]*, a direct comparison with the post-viral period would provide valuable data to support such a correlation.* **the writer mentions a limitation of the study** *(this was not addressed formally during the study)* **that leads to a suggestion for future research.**

**In Sentence 9** *V2D RNA is known to persist in the bodies of survivors long after disease onset*[30,42]*, and the results of our study provide compelling evidence that follow-up of all convalescent patients is warranted for at least 10 months after discharge from hospital.* **the writer mentions background factual information in the literature** *(is known to persist)* **to support the conclusions of the study.**

# Summary: Key of Discussion Model

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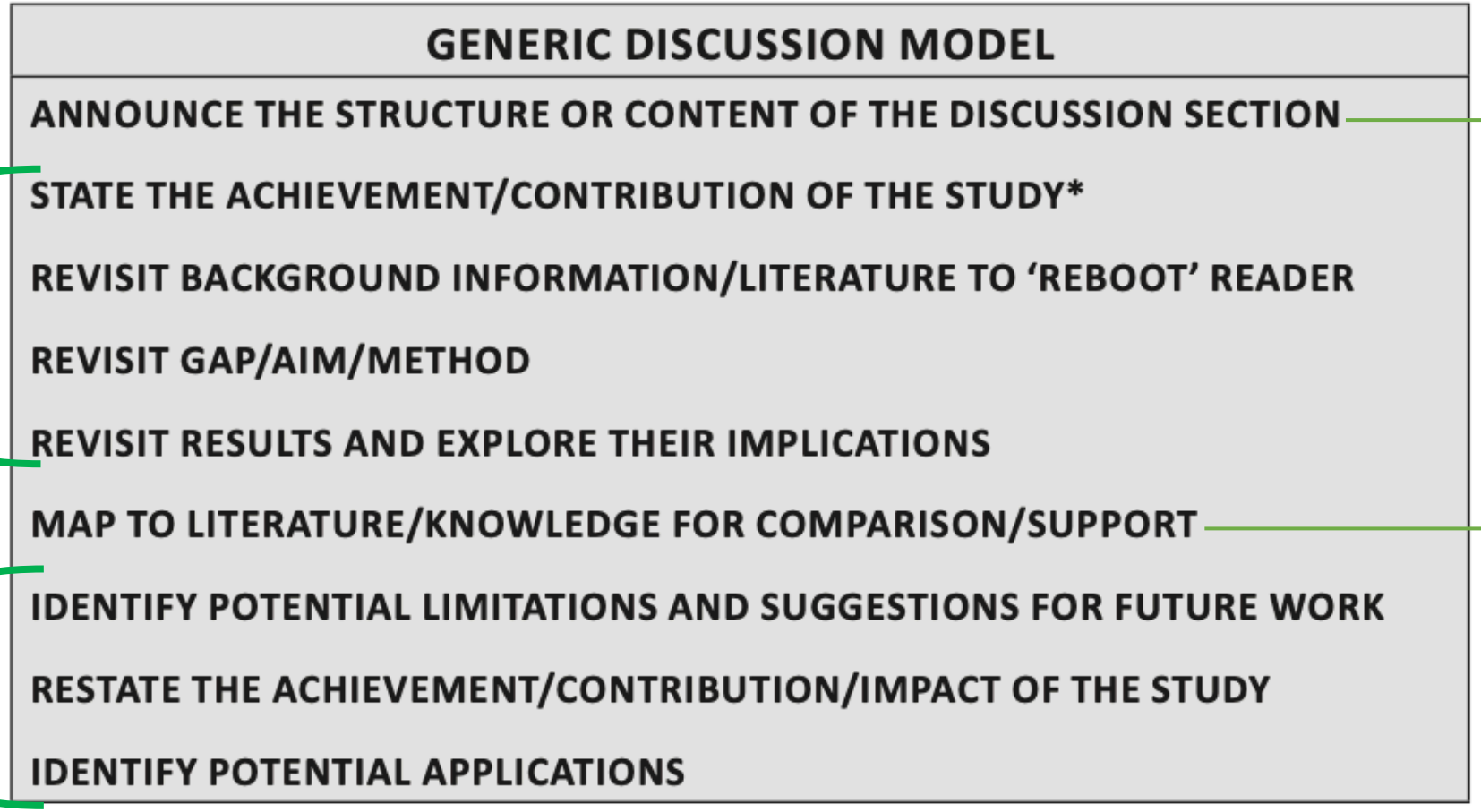
| | |
|---|---|
| In Sentence 1, the writer | explicitly identifies the contribution of the study in relation to the existing literature and knowledge. |
| In Sentence 2, the writer | identifies the specific novel feature of the study in relation to the existing literature, and summarises the results. |
| In Sentences 3 and 4, the writer | revisits the results and comments on them. |
| In Sentence 5, the writer | mentions a potential application arising from the results. |
| In Sentences 6, 7 and 8, the writer | mentions a limitation of the study that leads to a suggestion for future research. |
| In Sentence 9, the writer | mentions background factual information in the literature to support the conclusions of the study. |
| In Sentence 10, the writer | closes strongly with the applications of the study. |

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# Generic Discussion Model

# Generic Discussion Model

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You can present each hypothesis in bullet point

The main element of discussion

Only selected relevant literatures to strengthen your contribution

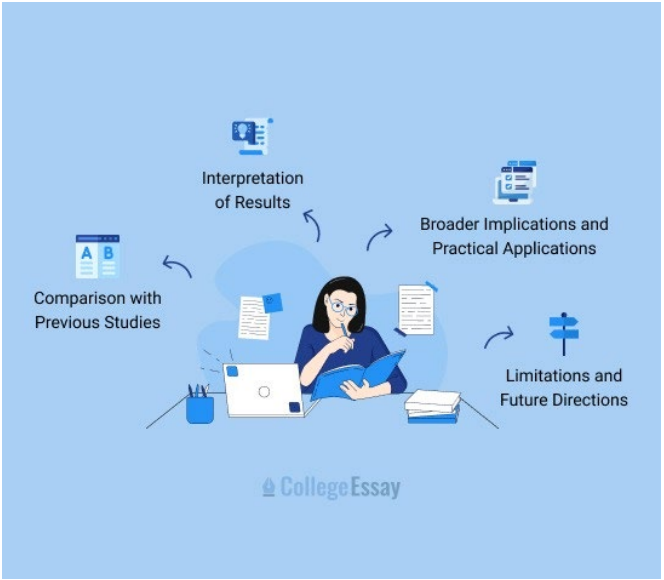
Happy words

4

# Useful Words and Phrases in Discussion

# Language Task

1. Map to literature/knowledge
2. Refine/explore implications
3. Achievement/contribution to literature/knowledge
4. Current and future work
5. Applications/use/applicability/implementation




https://collegeessay.org/

# 1. Map to literature/knowledge

| | |
|---|---|
| an alternative scheme/strategy | **Verbs:** |
| analogous to | This study/These results |
| comparable to | |
| consistent with | challenge |
| contrary to | compare well (with) |
| distinct from | complement |
| entirely different | confirm |
| equivalent to | conflict (with) |
| except for | contradict |
| fundamentally the same as | correspond to |
| identical to | corroborate |
| in accordance with | differ from |
| in agreement with | disprove |
| in conflict with | expand |
| in contrast to | extend |
| in good agreement (with) | improve |
| in line with | mirror |
| new/novel | modify |
| previously described/reported/suggested | pave the way for |
| rather than | prove |
| recently | provide insight into |
| significantly different (to/from) | provide support for |
| similar | refute |
| the first time/first of its kind | resemble |

| | |
|---|---|
| to the best of our knowledge, | shed new light on |
| unlike | substantiate |
| | support |
| | verify |

# Example usage

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- **Rather than** being excluded **as is often su**ggested, the sulfate reducers seemed to be thriving.
- The frequency domain algorithm proposed here **resembles** the so–called Frisch Scheme approach.
- **Unlike existing** control schemes, in this scheme the parameters can be freely changed during operation.
- This is **in agreement with** results from recent studies which suggest that iron reducers can modify their rate of respiration.
- Our data therefore **provide support for** the theory proposed by Stephen Robbins [Robbins *et al.,* 2013].
- The presented structure-based approach **complements existing** experimental methods.
- This study **extends** previous research by analysing corporate brand and industry image simultaneously.
- These results **are consistent with** the theory that fungal infection leads to plant death.
- The current study therefore both **confirmed and expanded** prior research into these mechanisms.
- On the basis of these results, **we challenge** the assumptions made by existing physical-layer security systems.

# 2. Refine/explore implications

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| | | |
|---|---|---|
| plausible<br>potential<br>tentative<br><br>to hypothesise<br>to postulate<br>to speculate<br>to theorise | modal verbs (esp. may/might/could)<br><br>it is conceivable that…<br>it is reasonable to assume that…<br>this is reinforced by…<br>this is substantiated by…<br>this points to…<br>we cannot rule out… | Clearly,<br>Indeed,<br>Perhaps |

## Example usage

1. **It is conceivable** that such a mechanism **could** play a role in many other disease states.
2. We **postulate that** Brownian motion of nanoparticles in nanofluids produces convectionlike effects at the nanoscale.
3. A **tentative** explanation is that the colloidal system may be agglomerated at that treatment level.
4. **We hypothesize that** our work could serve as a catalyst for new forms of real-time interventions.
5. It is intriguing **to speculate** that brown fat progenitors may also reside in peripheral nerves.
6. However, **we do not rule out** the possibility of a reaction with hemicellulose and lignin.

# 3. Achievement/contribution

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**(i) Positive language ('happy' words)**

| | | | | |
|---|---|---|---|---|
| accurate | efficient | reliable | **Verbs/verb phrases:** | prove |
| advantage | encouraging | robust | | provide a first step |
| appealing | entirely | significant | | |
| appropriate | exact | simple | allow | provide a framework |
| attractive | fast | smooth | avoid | |
| beneficial | favourable | stable | compare well with | provide evidence of |
| clear | feasible | straightforward | | |
| comprehensive | flexible | strong | confirm | provide insight into |
| convenient | important | successful | enable | |
| convincing | intuitive | superior | enhance | remove the need for |
| correct | low-cost | systematic | ensure | |
| cost-effective | new | unambiguous | explain | resolve |
| direct | novel | useful | facilitate | reveal |
| easy | practical | valid | help to | solve |
| economical | precise | valuable | improve | streamline |
| effective | productive | versatile | is able to | succeed in |
| | realistic | viable | offer | support |
| | relevant | | outperform | validate |
| | | | | yield |

**(ii) !-substitutes ('very happy' words)**

| | | | |
|---|---|---|---|
| compelling | exciting | powerful | undeniable |
| crucial | extraordinary | remarkable | unique |
| dramatic | ideal | superb | unusual |
| excellent | invaluable | surprising | unprecedented |
| exceptional | outstanding | striking | unquestionably |
| | perfect | | vital |

# Example usage

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- This results in a **cost-effective** approach which **significantly improves** scalability.
- We were able to demonstrate a **striking** difference between neonates at high and at low risk of atopy.
- This work demonstrates that multimodel predictions can **provide a more reliable estimate** of uncertainty.
- We describe not only neutral but also ionized systems with **unprecedented accuracy.**
- The system described here **enables fast and easy** analysis of LRRC8 proteins.
- The study **provides a first step** towards **a better** representation of southern African dust sources in dust emission models.

# 4. Current and Future Work

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| | | |
|---|---|---|
| a/the need for | holds promise | should be explored |
| at present | possible direction | should be investigated |
| currently | research opportunities include… | should be replicated |
| encouraging | | should be validated |
| fruitful | starting point | |
| promising | the next stage | further work is planned* |
| urgent | we recommend | future work/studies will* |
| | we suggest | is being investigated* |
| further work is needed | worthwhile | work is (currently) underway* |
| future work/ studies should | would be beneficial/ useful | work is in progress* |
| | would be of interest | |

**Note*: These indicate that the writer is currently working on this; they are NOT suggestions for research by others, or invitations to other researchers.

# Example usage

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- **Future studies should** investigate whether reducing household air pollution may lead to improvement in cardiac morbidity.
- Developing a model able to answer all these questions is an exciting challenge **for future studies**.
- **Future work should** focus on a qualitative analysis of patient-reported benefits of group therapy.
- **Further work is in progress** to determine whether a different drought response could help these trees to survive.
- In the future, **it would be of interest to** develop a more efficient numerical method for simulating these system dynamics.
- To support these findings, variations in cfDNA test performance **should be investigated further**.
- **Recommendations for future studies include** an investigation of polysaccharide redundancy during cell wall assembly.
- **Work is currently underway** to determine the extent of this genetic control.
- **Future research opportunities include** improving the measurement of sustainability indicators.

# 5. Applications/use/applicability/implementation

| | | |
|---|---|---|
| eventually | to apply | applicable |
| in due course | to enable | appropriate |
| in future | to facilitate | feasible |
| soon | to generalise | operable |
| | to generate | practicable |
| | to implement | practical |
| | to lead to | realistic |
| | to operate/put into operation | suitable |
| | to produce | viable |
| | to realise | |
| | to serve as | |
| | to use | |
| | to utilise | |

# Example usage

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- The study **provides** evidence-based guidance for **realistic** public health recommendations.
- The dissection protocol outlined here is **appropriate** for all long bone analyses including ex vivo imaging.
- The approach presented here is also **suitable** for problems where structural information is incomplete.
- The experimental paradigm outlined here could be **used** to evaluate future robotic social agents.
- The proposed technique could be **implemented** widely within the bioprocess industry, including in the production of antibiotics.
- Although it focuses on the specific case of supernovae, the concepts presented here are **applicable** to other astrophysical cases.
- Our protocol will **enable** the transport and storage of samples, thereby **facilitating** mitochondrial function analysis of liver biopsies.
- We see miR-CLIP as a **broadly applicable approach that can be used** in conjunction with others.
- This work shows that low-cost air quality sensor networks are **feasible for widespread use**.
- The data reported in this study could **eventually lead to** recommendations to guide optimal Clozapine use.

# Quiz: Modal Verbs

Match the modal verbs in Column A to their meanings in Column B. Most modal verbs can be used to communicate more than one meaning, so fill in the blank spaces with as many options as you can.

| COLUMN A | COLUMN B |
|---|---|
| 1 should<br>2 must<br>3 can<br>4 may<br>5 could<br>6 need to<br>7 might<br>8 have to | *The model **is able to** predict a wide range of experimental data.*<br>= The model _____ predict a wide range of experimental data.<br>***It is possible that** these interactions are the same for each species.*<br>= These interactions _____ be the same for each species.<br>*The ratio **is expected to** remain constant if the expansion is uniform.*<br>= The ratio _____ remain constant if the expansion is uniform.<br>***It is obvious that** this is the result of direct collision between the electron and the nucleus.*<br>= This _____ be the result of direct collision between the electron and the nucleus.<br>***It is advisable to** centrifuge the tubes before the experiment.*<br>= The tubes _____ be centrifuged before the experiment.<br>***It is necessary to** centrifuge the tubes before the experiment.*<br>= The tubes _____ be centrifuged before the experiment. |

# Answer!

*The model **is able to** predict a wide range of experimental data.*
The model **can/could** predict a wide range of experimental data.

***It is possible that** these interactions are the same for each species.*
These interactions **may/might/could** be the same for each species.

*The ratio **is expected to** remain constant if the expansion is uniform.*
The ratio **should** remain constant if the expansion is uniform.

***It is obvious that** this is the result of direct collision between the electron and the nucleus.*
This **must** be the result of direct collision between the electron and the nucleus.

***It is advisable to** centrifuge the tubes before the experiment.*
The tubes **should** be centrifuged before the experiment.

***It is necessary to** centrifuge the tubes before the experiment.*
The tubes **must/need to/have to** be centrifuged before the experiment.

“Writing in science is not only about reporting facts,
but about building understanding”

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